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STAFF REPORT FOR THE 2014 ARTICLE IV CONSULTATION—DEBT SUSTAINABILITYANALYSIS

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Prepared by the International Monetary Fund and the International Development Association.

Based on the new Low-Income Country Debt Sustainability Analysis (LIC DSA) framework, the updated DSA found that Ethiopia's risk of external debt distress remains low, but exposure to adverse shocks to exports shifts the risk to the borderline. The DSA indicates that Ethiopia's overall public sector debt dynamics is sustainable under the baseline scenario, but vulnerable under several alternative scenarios. Public sector debt ratios are projected to decline in the medium and long-term. The baseline scenario with a medium and long-term favorable trend in the main indicators, places the debt-to-exports ratio close to the threshold for a short period, with shocks to exports and borrowing conditions temporarily breaching the PV of debt-to-exports threshold. This underscores the need for the government to develop a strategy to contain the primary deficit as assumed in the baseline. A rapid increase in domestic public debt and non-concessional external borrowing could affect the risk assessment of Ethiopia. Maintaining the growth of exports through diversification of the export sector, developing a medium-term debt strategy for the public sector, and prudent non-concessional borrowing from external sources, remain crucial to maintaining a low risk of external debt distress.

BACKGROUND AND KEY FINDINGS

1. The last Debt Sustainability Analysis (DSA), prepared in August 2013, concluded that Ethiopia was at a low risk of external debt distress. Past reports explained that Ethiopia reached the completion point under the Heavily Indebted Poor Countries (HIPC) Initiative in 2004 and benefited from debt relief under the Multilateral Debt Relief Initiative (MDRI) in 2006.[1] In 2011/12,[2] public and publicly guaranteed (PPG) external debt declined to 18.4 percent of GDP, more than 5 percentage points relative to the previous year.[3] High GDP growth and negative domestic real interest rates on public debt contributed to the decline.

2. Previous DSA findings have been robust to the inclusion of remittances and higher non-concessional loans. Last year's external DSA did not result in any breach of indicative thresholds with or without remittances and debt-to-exports ratios were better in the baseline with those private transfers. The assessment was thus made not taking into account remittances, as recommended in the LIC DSA framework, and there was evidence of low risk of external debt distress. An alternative scenario and a general equilibrium model stretching the non-concessional borrowing for public investment on infrastructure confirmed the capacity of the country to absorb those resources.[4] Nonetheless, as Ethiopia reaches new debt levels, the assessment of the external DSA calls for the use of additional tools (as provided under the new Debt Sustainability Framework).

3. Ethiopia remains at low risk of external debt distress in 2014, but debt levels now warrant extended analysis of the incremental risk of higher debt and vulnerability of exports to terms of trade shocks and terms of loan. The present value (PV) of PPG external debt is expected to continue to increase from 14.1 percent of GDP in 2012/13 to 15.4 percent of GDP in 2013/14 under baseline scenario. The ratio of PV of PPG external debt to exports would rise from 110.6 percent to 115.0 percent in the same period, as a result of the additional buildup of external debt and low level of exports. The inclusion of remittances significantly lowers the baseline average of the debt-to-exports ratios in the projection period (2013/14–2033/34), but the results excluding and including workers' remittances are broadly the same.[5]

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[1] While Ethiopia has received debt relief from most of its creditors, it has not been able to reach agreement with bilateral official creditors from Bulgaria, Libya, and FR Yugoslavia and commercial creditors from Italy, former Czechoslovakia, India and FR Yugoslavia whose outstanding loans (US$382.9 million) accounted for 4.5 percent of the debt stock in 2012/13. HIPC terms are assumed for these loans. Negotiations with Russia on outstanding loans (US$161.6 million) have not completed. Thus, the debt service on these loans is assumed HIPC comparable treatment.

[2] The Ethiopian fiscal year runs from July 8 to July 7.

[3] PPG debt includes not only federal and regional governments' debt, but also encompasses all major state-owned enterprises, except Ethiopian Airlines (EAL). EAL debt is excluded from PPG debt, because, although owned by the government, it is run on commercial terms. EAL enjoys managerial independence, borrows without any government guarantees, publishes annual audited reports and has a sizeable profit margin.

[4] See Buffie et al (2014), 'Borrowing for Growth: Big pushes and Debt Sustainability in Low-income Countries', April (2014) and Box 1 in the main report.

[5] Based on the 2013 Country Policy and Institutional Assessment (CPIA) score (3.4), Ethiopia is classified as a medium performer. The thresholds for the debt burden for medium performers are 150, 40, and 250 for the PV of debt to

(continued)

There is a short-lived breach of the PV of debt-to-exports ratio under the exports and borrowing conditions shocks, that makes Ethiopia a borderline case under the new DSF and requires the use of additional country specific information to perform the DSA.

Table 1. Ethiopia: Comparison of PPG External Debt Baseline Scenario

Fiscal year ending July 7	2014	2015	2016	2017	2018	2019	2024	2034
			(Percent, unless otherwise indicated)					
			PV of Debt to Exports Ratio					
2014DSA	115.0	135.9	141.9	141.1	131.9	120.1	78.7	25.2
2013DSA	116.8	120.9	124.1	122.2	117.6	113.4	92.6	...
			PV of Debt to GDP Ratio					
2014DSA	15.4	18.9	21.0	21.9	21.3	20.4	15.3	6.7
2013DSA	16.4	17.5	18.8	19.1	19.0	18.9	16.2	...
			PV of Debt to Revenue Ratio					
2014DSA	102.0	131.5	147.1	154.6	149.4	140.8	105.4	38.7
2013DSA	127.9	131.9	141.9	144.6	144.7	140.8	121.9	...
			Debt Service to Exports Ratio					
2014DSA	7.7	9.0	8.8	8.7	8.4	8.6	7.7	2.7
2013DSA	6.4	7.2	7.1	7.5	8.1	8.4	8.2	...
Memorandum items:								
			Grant Element of New External Borrowing					
2014DSA	25.7	21.8	22.9	25.6	33.0	34.1	35.3	34.9
2013DSA	24.4	25.0	23.6	28.0	27.8	26.8	29.0	...
			New Commercial Loan Disbursements (billions of U.S. dollars)					
2014DSA	0.948	2.707	2.271	1.745	0.930	0.820	0.815	0.750
2013DSA	1.037	1.010	1.111	0.752	0.744	0.789	0.695	...
			Real GDP Growth (annual percent change)					
2014DSA	8.2	8.5	8.5	8.0	8.0	7.5	6.5	6.5
2013DSA	7.5	7.5	7.0	7.0	7.0	6.5	6.5	...
			Current Account Balance to GDP Ratio					
2014DSA	-7.1	-7.3	-6.7	-5.8	-5.5	-6.2	-4.0	-2.7
2013DSA	-5.7	-6.0	-5.8	-5.6	-5.2	-5.0	-4.4	...

Sources: Ethiopian authorities; IMF and World Bank staff estimates and projections.

exports, GDP, and revenue, respectively; a 10 percent band is applied to the thresholds to determine if the country is a borderline case; debt service thresholds are 20 and 30 percent of exports and revenue, respectively. In the scenarios that include workers' remittances, the corresponding threshold for PV of debt to exports and remittances is 120 percent, and is 16 percent for debt service to exports and remittances; the PV of debt to GDP and remittances is 36 percent.

4. The current DSA assumes a decline in the share of concessional loans and higher nonconcessional external loan disbursements between 2014/15 and 2019/20. Compared to the previous DSA, a greater increase in projected commercial loan disbursements for financing large investments by state-owned enterprises in the next five years is the main contributor to the buildup of new debt in the medium term. New commercial loans will result in subsequent increases in the various debt ratios and an increase in the grant element on new borrowing throughout the projection period. This DSA assumes disbursements of about US$1.7 billion in nonconcessional loans a year in the next five years, with Ethiotel (the telecommunications company) taking US$1.6 billion over three years. The DSA assumes that nonconcessional loan disbursements will remain at around US$745 million on average over the projection period (FY2018/19–FY2033/34). Over the time horizon of the DSA, 58 percent of new external loans are assumed to be concessional on average. Average maturity on all new external loans is assumed to be 26 years while new non-concessional loans are assumed to carry a maturity of 13–16 years. Average interest rates on new external loans are assumed at 2.2 percent over the horizon, and interest rates on new non-concessional loans are assumed to be in the 3–5 percent range.

5. In April 2013, IDA authorized a US$1 billion ceiling for FY12/13 and, in principle, FY13/14 and FY14/15 for Ethiopia. In June 2014, IDA concluded that Ethiopia had exceeded this limit by committing non concessional contracts totaling US$5.8 billion in FY12/13 and an estimated US$2.7 billion in FY13/14. In response, IDA converted the grant portion of the FY14/15 volume into regular credits and reduced Ethiopia's allocation for FY14/15 by 5 percent. Financing needs covered with non-concessional borrowing have been revised upwards from the previous DSA in view of the large cost of big projects in telecommunications, railways, roads, airport and sugar. Those projects have been identified as crucial for the success of the authorities' Growth and Transformation Plan (GTP). Frontloaded disbursements are associated with realistic pace of execution and the immediate needs for investment and related acquisitions in those sectors.

6. Commercial loans to finance large public investment projects by state-owned enterprises could increase the risk to Ethiopia's public debt sustainability. The state-owned power company, the Ethiopia Electricity Power Company, now divided into two entities for generation and distribution, is undertaking several large investment projects. Most rely on external assistance and loans (including both concessional and non-concessional), while the Renaissance Dam project, estimated by the authorities to cost 8.5 percent of 2013/14 GDP, is intended to be financed entirely domestically (except for the external non-concessional financing of the transmission line). The Ethiopian Railway Corporation signed contracts with Chinese and Turkish companies for projects whose total size is more than US$3.9 billion, or 7.8 percent of 2013/14 GDP. The telecommunication company signed two agreements with Chinese providers for US$1.6 billion suppliers' loans. As advised in previous DSAs, it would be prudent for the authorities to systematically update a medium-term debt management strategy (that includes SOEs) and to start monitoring the overall debt (external and domestic) of the consolidated public sector.

7. Authorities requested again the exclusion of the commercial external loans in the telecommunication sector from the DSA. While they agreed with the findings of the DSA, they specifically requested the exclusion of the commercial loans that Ethiotel has signed with Chinese equipment suppliers for a total of US$1.6 billion. The conditions of those loans are non-concessional and

the authorities explained that there is no government guarantee. The delivery of some of the equipment has been delayed and the authorities argued that capacity to absorb the resources and to put in place the equipment has been slow in the past. Contrary to the assumed parsimony in the previous DSA, staff found evidence for a frontloading of the disbursements, given the high relevance of the associated projects to achieve the goals under the GTP, and the fact that some projects have already been started. The authorities have also indicated that they intend to establish an agency to monitor and oversee the financial positions of all public enterprises and their overall impact on public sector finances.

MACROECONOMIC ASSUMPTIONS

8. The medium-term macroeconomic outlook incorporates an upward revision to output growth compared to the assumptions of the 2013 DSA (Box 1). Real GDP growth is forecasted at 8.2 percent in 2013/14 and 8.5 percent in the following two years, reflecting strong economic activity, mainly led by public infrastructure investment and favorable agricultural production. The projected long-run GDP growth rate is maintained at 6.5 percent.

9. Export growth is projected to recover from its weak performance in FY2012/13, while imports continue at a fast pace led by capital goods, fuel and other inputs. In the medium- to long-run, export growth would be supported by diversification of the export sector as Ethiopia explores new markets, consolidates the existing ones, and develops new export industries and markets new products, spurred by greater inflows of foreign direct investment, and domestic investments to boost service exports including electricity and international travel. Overall, exports of goods and services are projected to grow at 15 percent each year from 2014/15 to 2019/20.

EXTERNAL DEBT SUSTAINABILITY ANALYSIS

10. Under the baseline scenario (not including remittances), the PPG external debt indicators will rise in the next several years, but will remain under the relevant indicative thresholds (Figure 1). The PV of PPG external debt in percent of GDP would slightly increase in 2013/14 to 15.4 percent and is projected to reach 22.0 percent by 2016/17, reflecting the assumed steady increase in new loan disbursements, and subsequently would decline below 10 percent in the long run. The PV of debt in percent of exports reached 115.0 percent in 2013/14; and despite the expected recovery in exports, it is projected to continue increasing, peaking at 141.9 percent in 2015/16. The debt service-to-exports ratio remains well below the relevant threshold although it keeps rising to a peak of 9.0 percent in 2014/15, reflecting servicing of non-concessional loans by public enterprises.

11. Under the historical scenario, the debt stock indicators would be lower than under the baseline scenario. The scenario reflects significantly higher nominal GDP and export growth (than in the baseline), which helps to lower the debt ratios. It also reflects larger net debt creating flows (than in the baseline) that increases the debt ratios. The dynamic path under the historical scenario is determined by these two offsetting forces.

Box 1. Ethiopia: Macroeconomic Assumptions for the Baseline Scenario

Based on a partial financing of the Growth and Transformation Plan (GTP), real GDP growth is projected at 8.2 percent in 2013/14, 8.5 percent in the following two years, and to slightly tamp down to 8.0 percent until 2018/19 and to remain at 6.5 percent in the outer years.[1] This assessment contrasts with the government's growth projections in the GTP, and reflects remaining rigidities and constrains, given the limited space for private sector credit. CPI Inflation is projected to reach 8.5 percent by the end of 2013/14 and to stay at 9 percent in the long run. The primary balance of the public sector is projected to remain around -2.2 percent of GDP in average for the next five years, reflecting a substantial effort on tax collection and current expenditure.

The external current account deficit (before official transfers) is estimated at 9.8 percent in 2013/14. A further deterioration to 10.9 percent for next year will follow a slight improvement over the long-run to 6.8 percent in 2033/34.

Exports of goods and services are projected to grow by 12.5 percent in 2013/14, after a decline of 0.7 percent in 2012/13, and with an average growth of about 13.2 percent in the long run. A prolonged decline in commodity prices, especially in gold and coffee, largely offset gains in export volumes and other agricultural exports. Investments in targeted sectors that receive government support are expected to contribute to export growth, with export volume growth of about 14.7 percent for 2013/14 and about 13 percent in the next four years to just below 8 percent over the remaining DSA horizon. Imports of goods and services are projected to increase slightly above GDP as demand for imported capital goods is expected to grow fast during the GTP implementation. In the long run, imports are expected to grow in line with domestic output.

Workers' remittances have played a key role in supporting the balance of payments and in 2013/14 are expected to reach 8 percent of GDP. High growth, economic stability and the return of growth in developed countries is expected to maintain remittances at around 7 percent of GDP in the next five years and 4 percent in the long run.

Foreign direct investment (FDI) is projected increase from 2.7 percent of GDP in 2013/14 to 3.3 percent the following year and 3 to 5 percent in average in the medium and long term, on account of policies to promote greater foreign investment.

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[1] Domestic financing and prospective external inflows amount roughly to two third of full financing of the GTP.

12. The terms-of-trade shock is the most extreme potential risk for the sustainability of the external debt, breaching the threshold of PV of debt-to-exports ratio from 2015/16 to 2017/18 (without remittances). This makes the DSA for Ethiopia a borderline case. A potential deterioration in the terms-of-trade and a drastic change in the borrowing conditions could force a temporary breach of the PV of debt-to-exports ratio. The short-lived breach of just one indicator and the declining long-term trends indicate that Ethiopia is a special case that conforms to the borderline situation described in the most recent revision of the DSF. With other indicators well below the thresholds even for the most extreme shocks, the new DSA template provides for a further evaluation based in the probabilistic approach that takes into consideration historical growth performance. This complementary approach assesses the evolution of the probability of debt distress over time. In the case of Ethiopia, with high and sustained

economic growth, the additional evaluation indicates that the country's probability of debt distress is low even for the most extreme shocks (Figure 2).

PUBLIC DEBT SUSTAINABILITY ANALYSIS

13. Under the baseline scenario and similar to the findings in the 2013 DSA, the total public sector debt-to-GDP ratio would rise in the near term, but would start to decline in the medium- to long-term. This reflects large domestic borrowing and continued accumulation of external PPG debt by public enterprises to implement infrastructure investment projects contemplated in the GTP. It is expected that after an initial period of high spending, total public sector expenditure would revert to a lower level in the long run. Under the new DSF that imposes a threshold for the PV of Debt-to-GDP ratio, only the scenario that fixes the primary balance at a level in recent fiscal performance shows the ratio crossing the threshold with a growing trend. This is explained largely by the fact that Ethiopia has reported a higher primary deficit than in previous year, associated with the recent large investments that are part of the GTP. This is unlikely to remain unchanged in the next 20 years.

14. Debt stock related indicators are relatively flat and most extreme shocks do not alter significantly the results. All debt indicators show a mild increase in the medium-term, to return to levels similar to the 2013/14 ones. This result is predicated on the continuation of robust GDP growth, moderate public sector primary deficits, and most crucially - the authorities' policy of keeping domestic interest rates low. Real interest rates are assumed to remain in mildly negative territory with inflation expected to stay at 9 percent in the long run.

15. Public sector debt would grow in the long-run as a result of a permanent negative shock on GDP growth, but debt service-to-revenue could absorb this shock (Figure 3). The scenario with unchanged primary balance from 2013/14 shows particularly sharp deterioration because of the unusually large primary deficit in that particular year reflecting the high level of investment activity. The other two alternative scenarios (real GDP growth and primary balance at the historic average) permanently lower GDP growth and show trajectories with no important changes in the debt ratios during the DSA horizon.

16. The baseline scenario understates the public debt burden for the economy, because it reflects actual costs of borrowing by the public sectors, which remain significantly lower than inflation. Like in the previous DSA, with inflation projected to remain at a single-digit level, under current policy, interest rates on public enterprise domestic borrowing would not be at positive levels in real terms. Ethiopia's relatively benign public sector debt outlook hinges strongly on the continuation of the current interest rate policy. If the actual cost of borrowing were to rise above inflation, the debt indicators would worsen or fiscal adjustment would be required to maintain sustainability.

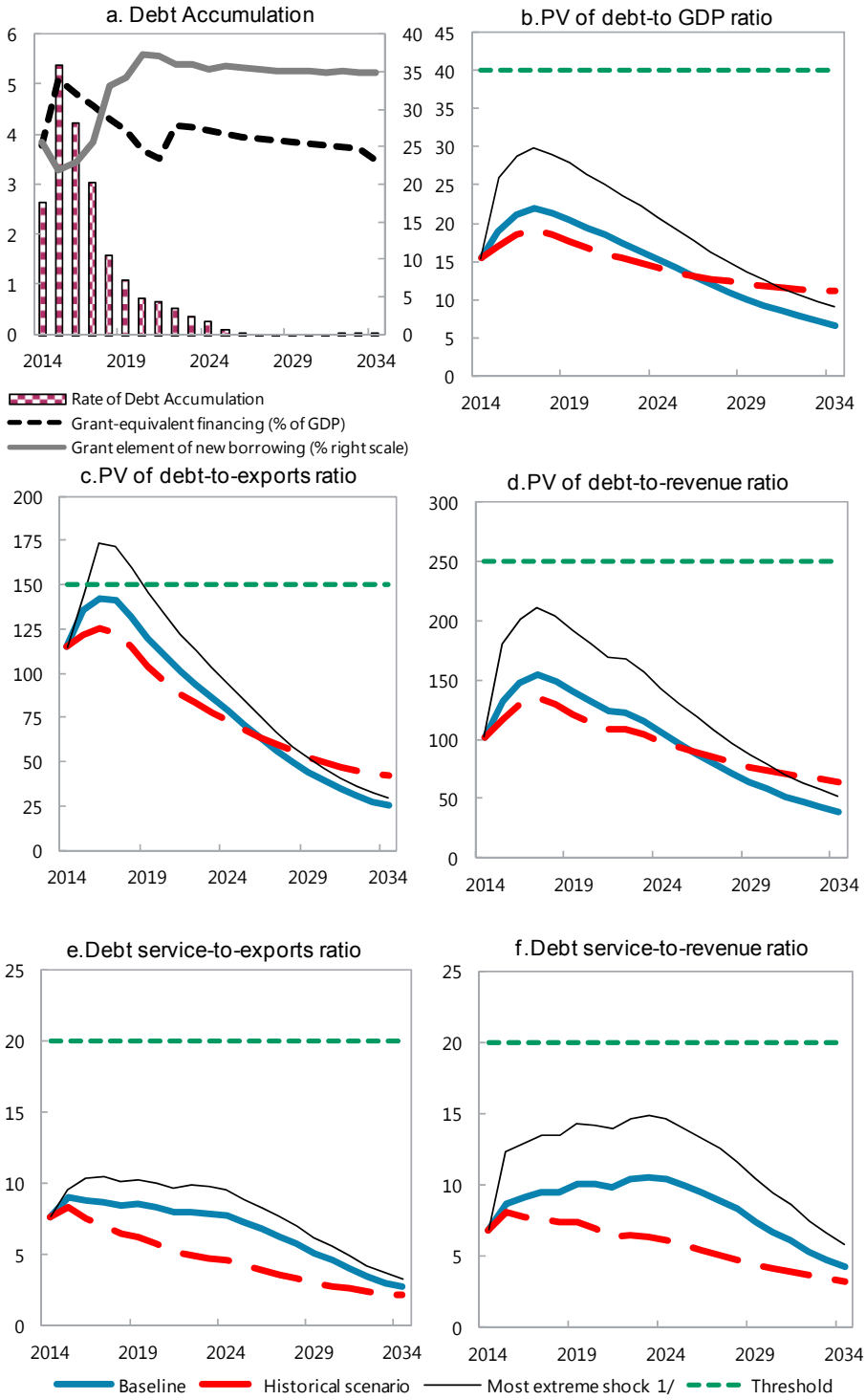
17. This assessment with a low risk of debt distress assumes disbursements of external borrowing lower than required for the GTP. The ongoing large public investment projects rely heavily on domestic financing and would lead to a large accumulation of public debt. Without an appropriate pacing of the public investment projects in the GTP, financing for the private sector could be squeezed. Monitoring the operations of the consolidated public sector, including contingent liabilities arising from financial transactions among public entities, is thus important.

CONCLUSION

18. The risk rating of Ethiopia's external and public debt distress remains low, but is on the cusp. Potential changes in the risk assessment can be triggered by a terms-of-trade shock or incurring higher than forecasted external debt as well as contracting debt on less favorable terms. The external debt ratios have been on a growing trend in recent years, and this trend is projected to continue over the medium-term. The results point to Ethiopia as a borderline case, with debt indicators and sensitivity analysis suggesting that the risk of debt distress may change from low to moderate. The exposure to a terms-of-trade shock is reflected in a short lived breach in the PV of debt-to-exports ratio under the most extreme shock. A milder terms-of-loans shock also breached this threshold for a short period.

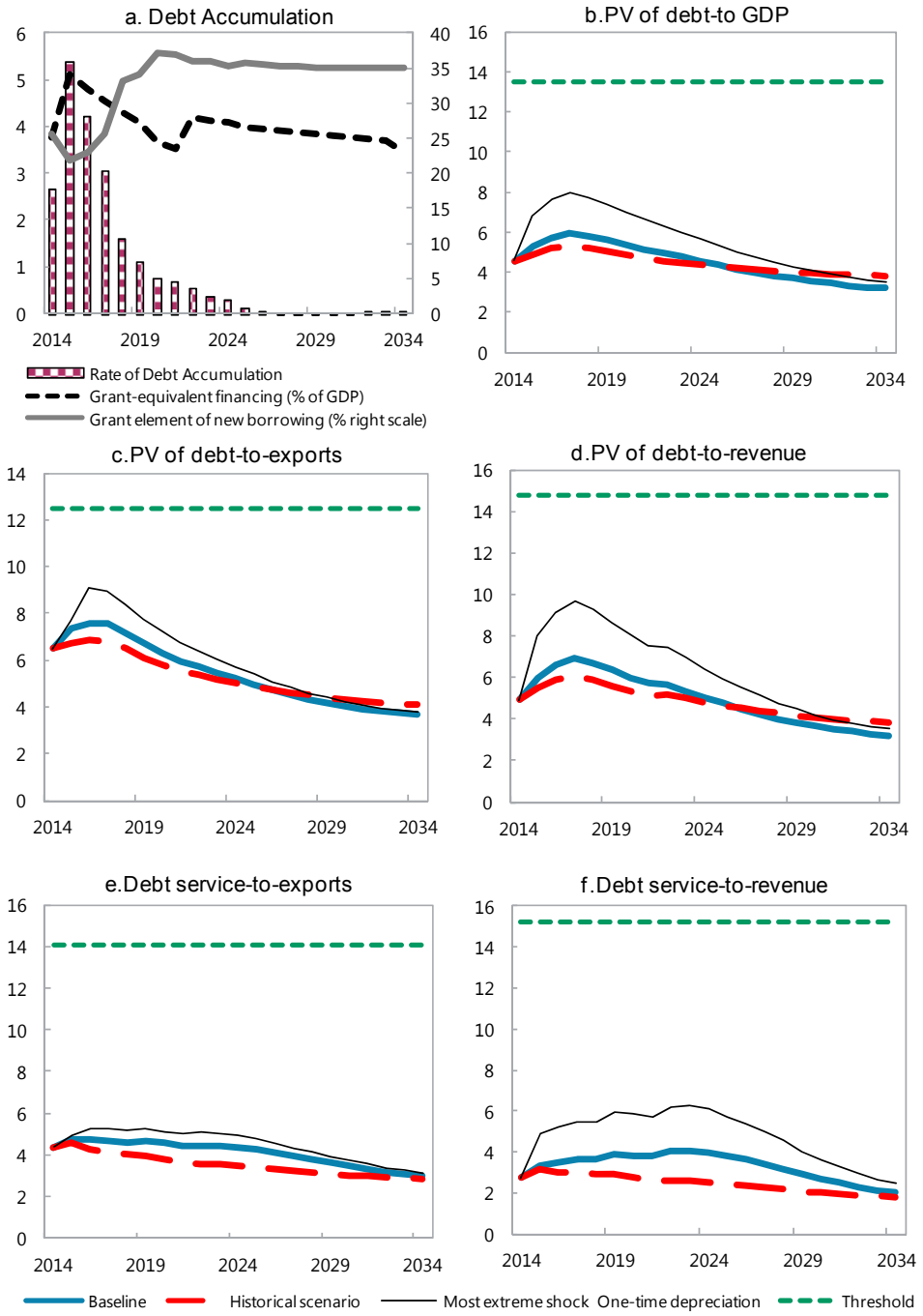
19. The Ethiopian authorities need to monitor closely the debt levels and the terms of the new loans, especially on commercial terms. The additional external debt intended to finance large projects on state own enterprises should be supported by higher exports. In this sense, ensuring the successful transition to a more competitive and diverse economy with a clear outward orientation is key to preserving the low risk of debt distress. The financing plan underlying the remaining GTP I and what is envisioned in the next GTP, should take into account these results. In addition to export diversification and improvements to competitiveness, particular emphasis should be placed on adequate concessionality of new external loans.

Figure 1. Ethiopia: Indicators of Public Guaranteed External Debt Under Alternative Scenarios, 2014–34[1]



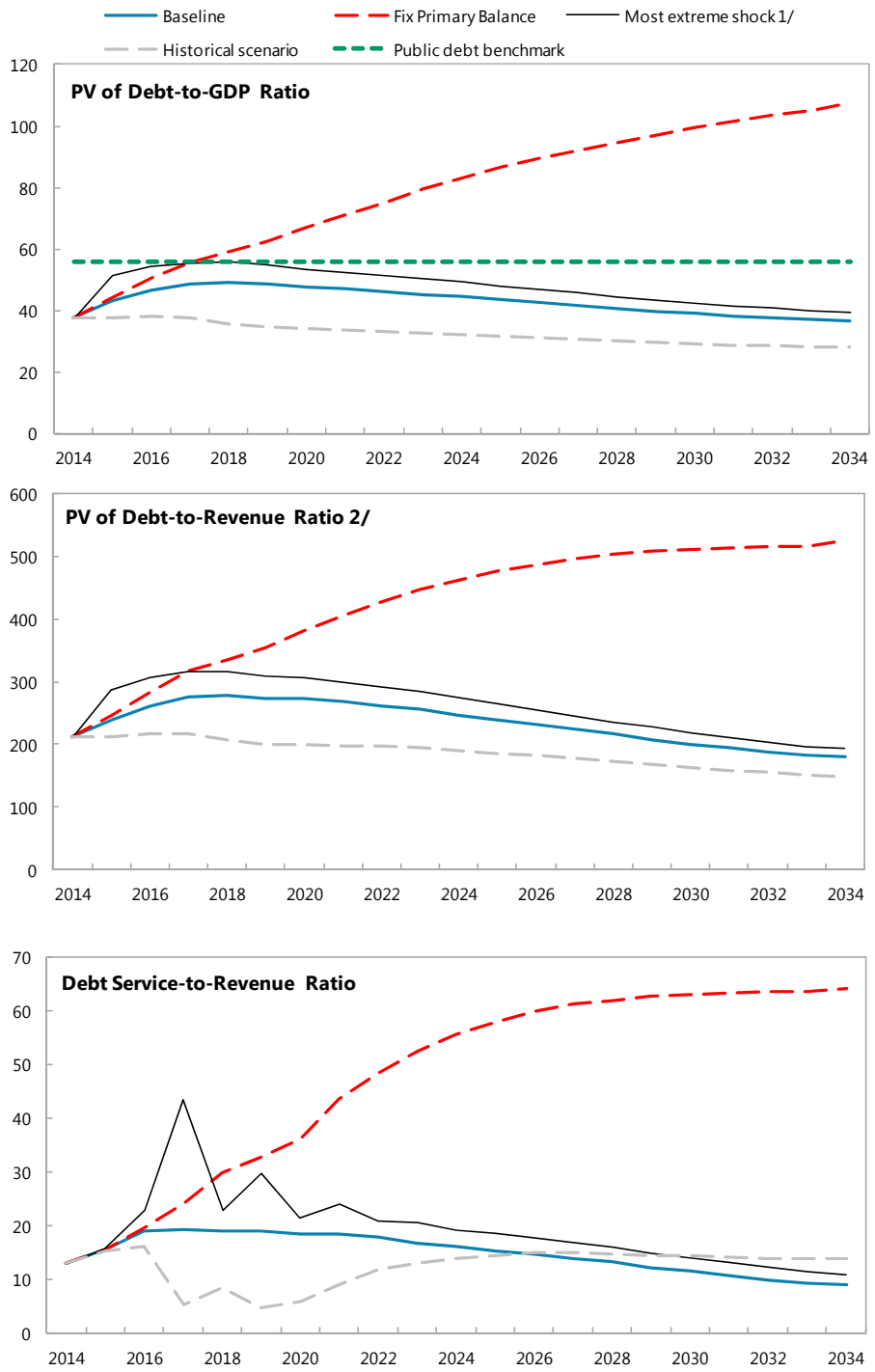
Sources: Ethiopian authorities; and IMF and World Bank staff estimates and projections.
1/ The most extreme stress test is the test that yields the highest ratio on or before 2024. In figure b. it corresponds to a One-time depreciation shock; in c. to a Exports shock; in d. to a One-time depreciation shock; in e. to a Exports shock and in figure f. to a One-time depreciation shock.

Figure 2. Ethiopia: Probability of Debt Distress of Public and Publicly Guaranteed External Debt under Alternative Scenarios, 2014–34[1]



Sources: Ethiopian authorities; and IMF and World Bank staff estimates and projections.
1/ The most extreme stress test is the test that yields the highest ratio on or before 2024. In figure b. it corresponds to a One-time depreciation shock; in c. to a Exports shock; in d. to a One-time depreciation shock; in e. to a Exports shock and in figure f. to a One-time depreciation shock.

Figure 3. Ethiopia: Indicators of Public Debt under Alternative Scenarios, 2014–34[1]



Sources: Ethiopian authorities; and IMF and World Bank staff estimates and projections.
1/ The most extreme stress test is the test that yields the highest ratio on or before 2024.
2/ Revenues are defined inclusive of grants.

Table 2. Ethiopia: External Debt Sustainability Framework, Baseline Scenario, 2011–34[1]

	Actual			Historical Average 6/	Standard Deviation 6/	Projections									
	2011	2012	2013			2014	2015	2016	2017	2018	2019	2014-2019 Average	2024	2034	2020-2034 Average
External debt (nominal) 1/	**26.1**	**20.6**	**24.5**			**27.3**	**31.4**	**33.7**	**34.5**	**33.5**	**32.2**		**25.2**	**14.1**	
of which: public and publicly guaranteed (PPG)	23.3	18.3	21.4			23.3	27.4	29.8	30.8	30.0	28.9		22.6	11.1	
Change in external debt	6.0	-5.5	4.0			2.8	4.1	2.3	0.8	-1.0	-1.4		-1.4	-0.7	
Identified net debt-creating flows	-4.6	-2.9	1.9			2.5	1.9	-0.4	-0.7	-1.1	-0.9		-0.9	-0.5	
Non-interest current account deficit	**0.5**	**6.3**	**5.8**	**4.7**	**2.6**	**6.7**	**6.8**	**6.0**	**5.1**	**4.8**	**5.4**		**3.5**	**2.3**	3.3
Deficit in balance of goods and services	15.1	18.0	17.0			17.4	18.2	17.0	15.5	14.9	15.4		11.9	8.3	
Exports	17.0	14.0	12.9			13.4	13.9	14.8	15.5	16.1	17.0		19.4	26.4	
Imports	32.1	32.0	29.9			30.8	32.1	31.8	31.0	31.0	32.4		31.3	34.7	
Net current transfers (negative = inflow)	-14.7	-11.8	-11.2	-13.8	1.4	-10.5	-11.4	-10.9	-10.3	-10.1	-9.9		-8.4	-5.9	-7.6
of which: official	-5.9	-4.2	-2.7			-2.7	-3.6	-3.5	-3.4	-3.4	-3.3		-3.5	-3.2	
Other current account flows (negative = net inflow)	0.0	0.0	0.0			-0.1	0.0	0.0	0.0	-0.1	-0.1		-0.1	0.0	
Net FDI (negative = inflow)	**-4.0**	**-2.5**	**-2.7**	**-2.6**	**0.8**	**-2.7**	**-3.3**	**-4.7**	**-4.1**	**-4.2**	**-4.8**		**-3.3**	**-2.3**	-3.2
Endogenous debt dynamics 2/	**-1.1**	**-6.7**	**-1.2**			**-1.5**	**-1.6**	**-1.8**	**-1.7**	**-1.7**	**-1.5**		**-1.0**	**-0.6**	
Contribution from nominal interest rate	0.2	0.2	0.3			0.4	0.5	0.7	0.7	0.8	0.8		0.6	0.3	
Contribution from real GDP growth	-2.1	-1.7	-1.8			-1.9	-2.1	-2.4	-2.5	-2.5	-2.3		-1.6	-0.9	
Contribution from price and exchange rate changes	0.9	-5.2	0.3			...	...	...	...	...	...		...	...	
Residual (3-4) 3/	**10.5**	**-2.6**	**2.1**			**0.2**	**2.2**	**2.7**	**1.5**	**0.1**	**-0.4**		**-0.5**	**-0.2**	
of which: exceptional financing	-0.4	-0.2	-0.1			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
PV of external debt 4/	...	...	17.2			19.4	22.9	24.9	25.6	24.8	23.7		17.9	9.7	
In percent of exports	...	...	133.6			144.7	164.6	168.2	165.1	153.7	139.4		92.2	36.6	
PV of PPG external debt	**...**	**...**	**14.1**			**15.4**	**18.9**	**21.0**	**21.9**	**21.3**	**20.4**		**15.3**	**6.7**	
In percent of exports	**...**	**...**	**109.1**			**115.0**	**135.9**	**141.9**	**141.1**	**131.9**	**120.1**		**78.7**	**25.2**	
In percent of government revenues	**...**	**...**	**90.0**			**102.0**	**131.5**	**147.1**	**154.6**	**149.4**	**140.8**		**105.4**	**38.7**	
Debt service-to-exports ratio (in percent)	**4.3**	**6.6**	**8.5**			**11.4**	**13.1**	**13.0**	**12.8**	**12.4**	**12.4**		**9.7**	**3.2**	
PPG debt service-to-exports ratio (in percent)	**2.9**	**4.9**	**6.6**			**7.7**	**9.0**	**8.8**	**8.7**	**8.4**	**8.6**		**7.7**	**2.7**	
PPG debt service-to-revenue ratio (in percent)	**3.7**	**5.0**	**5.4**			**6.8**	**8.7**	**9.1**	**9.5**	**9.5**	**10.1**		**10.3**	**4.2**	
Total gross financing need (Billions of U.S. dollars)	-0.8	2.0	2.0			2.8	2.9	2.0	2.0	1.9	2.2		2.6	2.7	
Non-interest current account deficit that stabilizes debt ratio	-5.5	11.8	1.8			4.0	2.7	3.7	4.4	5.8	6.8		4.9	3.1	
Key macroeconomic assumptions															
Real GDP growth (in percent)	11.4	8.8	9.7	10.9	1.1	8.2	8.5	8.5	8.0	8.0	7.5	8.1	6.5	6.5	6.7
GDP deflator in US dollar terms (change in percent)	-4.1	24.8	-1.6	7.5	12.5	-0.1	2.3	1.9	2.1	2.4	2.2	1.8	2.1	1.6	2.0
Effective interest rate (percent) 5/	1.1	1.2	1.4	1.0	0.3	1.7	2.2	2.3	2.4	2.5	2.5	2.3	2.4	2.3	2.3
Growth of exports of G&S (US dollar terms, in percent)	31.7	11.8	-0.7	18.3	10.0	12.5	15.1	17.9	15.5	15.0	15.7	15.3	11.2	9.5	12.1
Growth of imports of G&S (US dollar terms, in percent)	2.9	35.5	0.7	20.3	16.2	11.2	15.7	9.8	7.6	10.6	14.6	11.6	8.9	7.2	9.4
Grant element of new public sector borrowing (in percent)	...	...	...	...	...	25.7	21.8	22.9	25.6	33.0	34.1	27.2	35.3	34.9	35.5
Government revenues (excluding grants, in percent of GDP)	13.3	13.8	15.6			15.1	14.4	14.3	14.2	14.3	14.5		14.5	17.2	15.4
Aid flows (in Billions of US dollars) 7/	2.9	2.8	2.4			2.3	2.9	3.1	3.3	3.5	3.6		5.4	10.3	
of which: Grants	1.9	1.8	1.2			1.3	2.0	2.1	2.3	2.5	2.7		4.4	9.4	
of which: Concessional loans	1.0	1.0	1.2			1.0	1.0	1.0	1.0	1.0	0.9		1.0	1.0	
Grant-equivalent financing (in percent of GDP) 8/	...	...	...			3.8	5.1	4.8	4.6	4.3	4.1		4.1	3.5	3.8
Grant-equivalent financing (in percent of external financing) 8/	...	...	...			54.1	48.2	52.5	58.5	69.6	72.3		79.6	88.3	82.6
Memorandum items:															
Nominal GDP (Billions of US dollars)	31.4	42.6	46.0			49.7	55.1	61.0	67.3	74.4	81.8		125.6	291.0	
Nominal dollar GDP growth	6.8	35.9	7.9			8.1	10.9	10.6	10.3	10.6	9.9	10.1	8.8	8.2	8.8
PV of PPG external debt (in Billions of US dollars)			6.2			7.4	10.1	12.4	14.3	15.3	16.1		18.6	18.8	
(PVt-PVt-1)/GDPt-1 (in percent)						2.6	**5.4**	4.2	3.0	1.6	1.1	3.0	0.3	0.0	0.2
Gross workers' remittances (Billions of US dollars)	1.9	1.9	2.5			2.5	2.6	2.7	2.8	3.0	4.0		4.0	4.0	
PV of PPG external debt (in percent of GDP + remittances)	...	...	13.3			14.7	18.1	20.1	21.0	20.5	19.5		14.8	6.6	
PV of PPG external debt (in percent of exports + remittances)	...	...	76.8			83.7	101.9	109.1	110.9	105.7	93.1		67.5	24.0	
Debt service of PPG external debt (in percent of exports + remittances)	...	...	4.6			5.6	6.7	6.8	6.8	6.7	6.7		6.6	2.6	

Sources: Ethiopian authorities; and IMF and World Bank staff estimates and projections.

1/ Includes both public and private sector external debt.

2/ Derived as [r - g - ρ(1+g)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.

3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.

4/ Assumes that PV of private sector debt is equivalent to its face value.

5/ Current-year interest payments divided by previous period debt stock.

6/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

7/ Defined as grants, concessional loans, and debt relief.

8/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

Table 3. Ethiopia: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2014–34

	Projections							
	2014	2015	2016	2017	2018	2019	**2024**	2034
PV of debt-to GDP ratio								
Baseline	15	19	21	22	21	20	**15**	7
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2014-2034 1/	15	17	19	19	19	18	**14**	11
A2. New public sector loans on less favorable terms in 2014-2034 2	15	19	23	25	25	25	**21**	13
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2015-2016	15	18	20	21	20	19	**14**	6
B2. Export value growth at historical average minus one standard deviation in 2015-2016 3/	15	19	22	23	22	21	**16**	7
B3. US dollar GDP deflator at historical average minus one standard deviation in 2015-2016	15	20	24	24	24	23	**17**	7
B4. Net non-debt creating flows at historical average minus one standard deviation in 2015-2016 4/	15	19	22	22	22	21	**15**	7
B5. Combination of B1-B4 using one-half standard deviation shocks	15	18	20	21	20	20	**15**	6
B6. One-time 30 percent nominal depreciation relative to the baseline in 2015 5/	15	26	29	30	29	28	**21**	9
PV of debt-to-exports ratio								
Baseline	115	136	142	141	132	120	**79**	25
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2014-2034 1/	115	121	125	123	115	104	**73**	42
A2. New public sector loans on less favorable terms in 2014-2034 2	115	139	154	159	154	144	**109**	50
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2015-2016	115	131	137	136	127	116	**76**	24
B2. Export value growth at historical average minus one standard deviation in 2015-2016 3/	115	144	174	172	160	145	**94**	29
B3. US dollar GDP deflator at historical average minus one standard deviation in 2015-2016	115	131	137	136	127	116	**76**	24
B4. Net non-debt creating flows at historical average minus one standard deviation in 2015-2016 4/	115	134	146	145	135	123	**79**	25
B5. Combination of B1-B4 using one-half standard deviation shocks	115	131	147	146	136	124	**81**	26
B6. One-time 30 percent nominal depreciation relative to the baseline in 2015 5/	115	131	137	136	127	116	**76**	24
PV of debt-to-revenue ratio								
Baseline	102	132	147	155	149	141	**105**	39
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2014-2034 1/	102	117	130	135	130	122	**98**	64
A2. New public sector loans on less favorable terms in 2014-2034 2	102	135	159	174	175	169	**146**	76
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2015-2016	102	126	139	146	141	132	**99**	36
B2. Export value growth at historical average minus one standard deviation in 2015-2016 3/	102	131	156	163	157	147	**109**	39
B3. US dollar GDP deflator at historical average minus one standard deviation in 2015-2016	102	137	164	173	167	157	**117**	43
B4. Net non-debt creating flows at historical average minus one standard deviation in 2015-2016 4/	102	130	152	158	153	144	**106**	38
B5. Combination of B1-B4 using one-half standard deviation shocks	102	124	142	149	144	135	**101**	37
B6. One-time 30 percent nominal depreciation relative to the baseline in 2015 5/	102	180	201	211	204	192	**143**	52

Table 3. Ethiopia: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2014–34 (concluded)

(percent)

Debt service-to-exports ratio								
Baseline	8	9	9	9	8	9	**8**	3
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2014-2034 1/	8	8	7	7	6	6	**5**	2
A2. New public sector loans on less favorable terms in 2014-2034 2	8	9	8	9	9	8	**9**	4
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2015-2016	8	9	9	9	8	9	**8**	3
B2. Export value growth at historical average minus one standard deviation in 2015-2016 3/	8	10	10	10	10	10	**9**	3
B3. US dollar GDP deflator at historical average minus one standard deviation in 2015-2016	8	9	9	9	8	9	**8**	3
B4. Net non-debt creating flows at historical average minus one standard deviation in 2015-2016 4/	8	9	9	9	9	9	**8**	3
B5. Combination of B1-B4 using one-half standard deviation shocks	8	9	9	9	9	9	**8**	3
B6. One-time 30 percent nominal depreciation relative to the baseline in 2015 5/	8	9	9	9	8	9	**8**	3
Debt service-to-revenue ratio								
Baseline	7	9	9	10	10	10	**10**	4
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2014-2034 1/	7	8	8	8	7	7	**6**	3
A2. New public sector loans on less favorable terms in 2014-2034 2	7	9	8	10	10	10	**12**	7
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2015-2016	7	9	9	9	9	10	**10**	4
B2. Export value growth at historical average minus one standard deviation in 2015-2016 3/	7	9	9	10	10	10	**11**	4
B3. US dollar GDP deflator at historical average minus one standard deviation in 2015-2016	7	9	11	11	11	12	**12**	5
B4. Net non-debt creating flows at historical average minus one standard deviation in 2015-2016 4/	7	9	9	10	10	10	**11**	4
B5. Combination of B1-B4 using one-half standard deviation shocks	7	9	9	9	9	10	**10**	4
B6. One-time 30 percent nominal depreciation relative to the baseline in 2015 5/	7	12	13	13	13	14	**15**	6
Memorandum item:								
Grant element assumed on residual financing (i.e., financing required above baseline) 6/	29	29	29	29	29	29	**29**	29

Sources: Ethiopian authorities; and IMF and World Bank staff estimates and projections.

1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
2/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the baseline.
3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly assuming an offsetting adjustment in import levels).
4/ Includes official and private transfers and FDI.
5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.
6/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.

Table 4. Ethiopia: Public Sector Debt Sustainability Framework, Baseline Scenario, 2011–34

(percent of GDP, unless otherwise indicated)

	Actual					Estimate	Projections								
	2011	2012	2013	Average 5/	Standard Deviation 5/	2014	2015	2016	2017	2018	2019	2014-19 Average	2024	2034	2020-34 Average
Public sector debt 1/	38.9	33.2	38.3			45.5	51.5	55.3	57.3	57.8	57.1		51.8	41.2	
of which: foreign-currency denominated	23.3	18.3	21.4			23.3	27.4	29.8	30.8	30.0	28.9		22.6	11.1	
Change in public sector debt	-1.5	-5.7	5.1			7.2	6.0	3.8	2.1	0.5	-0.7		-1.2	-0.5	
Identified debt-creating flows	-3.6	-7.4	2.5			7.0	4.2	1.4	0.9	0.7	-1.3		-2.4	-0.8	
Primary deficit	2.5	3.7	4.6	3.3	1.7	10.0	8.7	6.3	6.0	6.2	4.1	6.9	2.2	3.0	2.6
Revenue and grants	19.3	17.9	18.3			17.8	18.0	17.8	17.6	17.6	17.8		18.0	20.5	
of which: grants	5.9	4.2	2.7			2.7	3.6	3.5	3.4	3.4	3.3		3.5	3.2	
Primary (noninterest) expenditure	21.7	21.7	22.9			27.7	26.6	24.1	23.6	23.8	21.8		20.2	23.4	
Automatic debt dynamics	-5.8	-10.7	-1.9			-3.0	-4.5	-4.9	-5.1	-5.5	-5.4		-4.6	-3.8	
Contribution from interest rate/growth differential	-7.1	-6.6	-2.9			-3.2	-4.4	-4.9	-5.1	-5.4	-5.3		-4.5	-3.8	
of which: contribution from average real interest rate	-3.0	-3.5	0.0			-0.3	-0.9	-0.9	-1.0	-1.1	-1.2		-1.3	-1.2	
of which: contribution from real GDP growth	-4.1	-3.2	-2.9			-2.9	-3.5	-4.0	-4.1	-4.3	-4.0		-3.2	-2.6	
Contribution from real exchange rate depreciation	1.3	-4.1	1.0			0.2	-0.1	0.0	-0.1	-0.1	-0.1		...	...	
Other identified debt-creating flows	-0.3	-0.4	-0.1			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Privatization receipts (negative)	-0.3	-0.4	-0.1			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Recognition of implicit or contingent liabilities	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Debt relief (HIPC and other)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Other (specify, e.g. bank recapitalization)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Residual, including asset changes	2.2	1.6	2.6			0.2	1.8	2.4	1.2	-0.2	0.6		1.2	0.3	
Other Sustainability Indicators															
PV of public sector debt	...	...	30.9			37.6	43.0	46.5	48.5	49.1	48.7		44.5	36.8	
of which: foreign-currency denominated	...	...	14.1			15.4	18.9	21.0	21.9	21.3	20.4		15.3	6.7	
of which: external	...	...	14.1			15.4	18.9	21.0	21.9	21.3	20.4		15.3	6.7	
PV of contingent liabilities (not included in public sector debt)	...	...	...			...	...	...	...	...	...		...	...	
Gross financing need 2/	4.2	5.5	6.8			12.3	11.5	9.7	9.4	9.5	7.4		5.1	4.8	
PV of public sector debt-to-revenue and grants ratio (in percent)	...	...	168.8			211.5	239.4	260.7	275.0	278.6	273.9		247.2	180.0	
PV of public sector debt-to-revenue ratio (in percent)	...	...	198.1			248.7	299.0	324.8	341.9	344.7	335.7		307.0	213.5	
of which: external 3/	...	...	90.0			102.0	131.5	147.1	154.6	149.4	140.8		105.4	38.7	
Debt service-to-revenue and grants ratio (in percent) 4/	8.9	10.2	12.2			12.9	15.7	18.9	19.4	19.0	18.9		16.1	8.9	
Debt service-to-revenue ratio (in percent) 4/	12.9	13.3	14.3			15.2	19.6	23.6	24.1	23.5	23.2		20.0	10.5	
Primary deficit that stabilizes the debt-to-GDP ratio	3.9	9.5	-0.5			2.8	2.6	2.5	4.0	5.7	4.7		3.4	3.5	
Key macroeconomic and fiscal assumptions															
Real GDP growth (in percent)	11.4	8.8	9.7	10.9	1.1	8.2	8.5	8.5	8.0	8.0	7.5	8.1	6.5	6.5	6.7
Average nominal interest rate on forex debt (in percent)	0.8	1.1	1.4	0.8	0.3	1.7	2.1	2.2	2.3	2.4	2.3	2.2	2.2	1.7	2.0
Average real interest rate on domestic debt (in percent)	-13.6	-21.4	1.2	-10.2	8.1	-1.8	-4.3	-4.3	-4.6	-5.0	-5.2	-4.2	-4.9	-4.4	-4.8
Real exchange rate depreciation (in percent, + indicates depreciation)	7.6	-19.4	6.2	-3.2	11.6	0.8	...	...	...	...	...	...	...	...	...
Inflation rate (GDP deflator, in percent)	19.8	34.2	5.3	15.8	10.7	7.8	9.5	9.1	9.3	9.5	9.4	9.1	9.3	8.7	9.2
Growth of real primary spending (deflated by GDP deflator, in percent)	15.9	8.5	15.8	4.1	6.7	31.1	4.0	-1.6	5.9	8.7	-1.4	7.8	7.3	7.4	7.2
Grant element of new external borrowing (in percent)	...	...	...	...	...	25.7	21.8	22.9	25.6	33.0	34.1	27.2	35.3	34.9	...

Sources: Ethiopian authorities; and IMF and World Bank staff estimates and projections.

1/ Public sector includes general government and state own enterprises (Excluding Ethiopian Airlines).

2/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.

3/ Revenues excluding grants.

4/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.

5/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

Table 5. Ethiopia: Sensitivity Analysis for Key Indicators of Public Debt, 2014–34

(percent)

	Projections							
	2014	2015	2016	2017	2018	2019	2024	2034
PV of Debt-to-GDP Ratio								
Baseline	38	43	46	48	49	49	44	37
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	38	38	38	38	36	35	32	28
A2. Primary balance is unchanged from 2014	38	44	51	56	59	63	83	107
A3. Permanently lower GDP growth 1/	38	43	47	49	50	50	47	43
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2015-2016	38	42	45	47	47	46	41	32
B2. Primary balance is at historical average minus one standard deviations in 2015-2016	38	40	42	45	46	46	42	35
B3. Combination of B1-B2 using one half standard deviation shocks	38	39	40	42	42	42	37	29
B4. One-time 30 percent real depreciation in 2015	38	49	52	54	54	53	49	42
B5. 10 percent of GDP increase in other debt-creating flows in 2015	38	51	54	56	56	55	49	40
PV of Debt-to-Revenue Ratio 2/								
Baseline	211	239	261	275	279	274	247	180
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	211	211	216	216	207	200	190	149
A2. Primary balance is unchanged from 2014	211	246	284	316	335	354	462	525
A3. Permanently lower GDP growth 1/	211	240	262	277	282	279	259	211
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2015-2016	211	236	253	266	268	262	230	158
B2. Primary balance is at historical average minus one standard deviations in 2015-2016	211	222	238	255	260	257	234	173
B3. Combination of B1-B2 using one half standard deviation shocks	211	216	225	239	242	238	209	141
B4. One-time 30 percent real depreciation in 2015	211	274	291	304	306	301	275	205
B5. 10 percent of GDP increase in other debt-creating flows in 2015	211	286	306	315	316	308	274	194
Debt Service-to-Revenue Ratio 2/								
Baseline	13	16	19	19	19	19	16	9
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	13	15	16	5	8	5	14	14
A2. Primary balance is unchanged from 2014	13	16	19	24	30	33	55	64
A3. Permanently lower GDP growth 1/	13	16	19	20	19	19	18	13
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2015-2016	13	16	18	18	18	17	14	6
B2. Primary balance is at historical average minus one standard deviations in 2015-2016	13	16	18	10	15	15	14	8
B3. Combination of B1-B2 using one half standard deviation shocks	13	16	17	8	12	12	11	3
B4. One-time 30 percent real depreciation in 2015	13	17	23	24	24	25	24	15
B5. 10 percent of GDP increase in other debt-creating flows in 2015	13	16	23	43	23	30	19	11

Sources: Ethiopian authorities; and IMF and World Bank staff estimates and projections.

1/ Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of the length of the projection period.

2/ Revenues are defined inclusive of grants.